

Methodology — Presenter Script & Q&A Prep

1. Presentation Script

Design & data

Our study is a **retrospective cohort study**, using the MIMIC-IV database from PhysioNet.

In simple terms, **MIMIC-IV** is a large, de-identified database of real ICU patient records, open to approved researchers. From it, we used a dataset that gives hourly ventilator readings for each patient.

Population (see flow chart)

We started with about **110 million hourly rows, from 50,920 ICU patients**. Most of those rows were empty because the dataset fills every patient out to 90 days, whether they were there that long or not. Once we cut each patient off at their real death or discharge, that removed about **97 million padding rows**, leaving roughly **12.7 million genuine hourly records**. Then we kept only patients for whom we could actually compute mechanical power — that excluded more than **31,800 patients** who never had the values we needed. After quality filtering and collapsing each patient into one summary row, our final cohort was **18,980 mechanically ventilated adults**.

And they needed at least **four hours** of usable data in the first 24 hours. The four-hour minimum makes sure we captured steady, real ventilation — not brief setup settings, or patients taken off the ventilator almost right away.

How did we calculate mechanical power? For each hour, we used a standard published equation, based on the patient's routine ventilator settings. Because the equation uses **peak** airway pressure, it needs no special bedside measurement — so we could calculate it from data that was already recorded. We did this for the first 24 hours.

We measured this exposure in three ways. Our main one was the **time-weighted average**: since settings are recorded only when they *change*, a simple average could mislead, so we weighted each setting by how long it was actually used. Our outcome was **death within 28 days**, measured from the moment of intubation. We adjusted for factors such as age, sex, oxygenation, and signs of shock, and used logistic regression with pre-planned subgroups. The data work was done in SQL Server, and the statistics in SPSS.

So that's the cohort and the analysis — and my colleague will now take you through what we actually found.

2. Delivery Notes

- Open on the number. Pause slightly after “one hundred and ten million” — let the scale land. That's your hook.
- Point at the flow chart through the entire funnel paragraph, tracking down it stage by stage.

- Stress four numbers as distinct beats, with a small pause after each: 50,920 → ~97 million removed → ~12.7 million → 18,980. Tap the final box when you say 18,980.
- Land the closing arc slowly: “50,920 patients down to 18,980” is the payoff — let it breathe.
- Say “empty padding” with a slight lift — it’s the counterintuitive idea, so don’t rush past it.
- Keep the mechanical-power paragraph brisk — a quick nod toward your intro teammate signals you’re deliberately not repeating the concept.
- Slow down a touch on “peak pressure means no special maneuver” — it’s the one technical insight worth letting sink in.
- Hand-off: turn toward your results teammate as you say “my colleague” — make the transition visually intentional.
- In your read-copy, spell out the big numbers (“fifty thousand nine hundred twenty,” “just under nineteen thousand”) to avoid stumbling on stage.

3. Q&A Preparation

Verified clean against the manuscript by two independent fact-checkers. Answers are written to say out loud.

1) MIMIC & the data source

Q. What exactly is MIMIC-IV? *[basic]*

A. Medical Information Mart for Intensive Care, version 4. A large, freely accessible research database of real ICU records — essentially an anonymized export of EHRs from Beth Israel Deaconess Medical Center in Boston, built by MIT’s Lab for Computational Physiology and distributed through PhysioNet. We used version 3.1.

Q. Why a US single-center database, not local data? *[intermediate]*

A. We needed hourly ventilator settings on tens of thousands of patients; that scale and depth isn’t openly available locally. MIMIC is one of the most validated, widely used critical-care databases in the world, which makes our work transparent and reproducible. Being single-center and US-based is a generalizability limitation we acknowledge — external validation is a sensible next step.

Q. What does “de-identified” mean for ethics? *[intermediate]*

A. All protected health info is stripped — no names, no record numbers, dates shifted — so a record can’t be traced to a real person. That’s what lets it be shared for research under a strict data use agreement.

Q. How did you get access? What’s the CITI training? *[intermediate]*

A. Access is credentialed, not open download. One team member completed PhysioNet credentialing, which requires the CITI “Data or Specimens Only Research” ethics course. After signing the data use agreement, no identifiable data was ever shared.

Q. Why no King Abdulaziz ethics approval? *[tough]*

A. It’s a fully de-identified, publicly accessible dataset with its own ethics governance (PhysioNet’s agreement + CITI training). There are no identifiable human subjects on our end, so a secondary analysis needs no additional institutional approval.

Q. What’s the “derived dataset” and how did you link it? *[tough]*

A. A published PhysioNet dataset, “A Temporal Dataset for Respiratory Support in Critically Ill Patients” (v1.1.0), derived from MIMIC-IV, giving hourly ventilator settings, respiratory parameters, and blood gases. We linked it back to the core MIMIC tables using `subject_id` (patient) and `hadm_id` (admission) to confirm intubation and pull demographics, comorbidities, and vital status.

2) The cohort funnel & numbers

Q. 110 million rows but only 18,980 patients — what were those rows? *[basic]*

A. Those are hourly observations, not patients. The dataset forward-fills every patient out to 90 days regardless of real ICU stay, so most rows are empty padding after death/discharge. Truncating each patient at their real terminal event removed ~97 million padding rows, leaving ~12.7 million genuine records — and that step removed zero patients; we only deleted fictitious time.

Q. Why did patients fall from 50,920 to 18,980? *[basic]*

A. Because not every ICU patient is ventilated, and our study is specifically about ventilated patients. We excluded ~31,800 with no computable mechanical power (never invasively ventilated, or never had the four settings recorded), plus a handful from QC and final aggregation. It’s not data loss — it’s the funnel narrowing to the ventilated population we’re studying.

Q. Isn’t the 4-hour cutoff arbitrary? *[intermediate]*

A. It’s a pragmatic floor, not a magic number. It captures a sustained ventilation strategy rather than first-hour setup noise, and it drops patients who died or were extubated almost immediately, where there isn’t enough exposure to characterize. It trades a little sample for a much cleaner exposure window.

Q. How did you confirm invasive ventilation? *[intermediate]*

A. We didn’t trust the dataset’s native mode marker — it was unreliable. We anchored to real airway events: intubation/extubation timestamps in the MIMIC procedures table (itemid 225792, “Invasive Ventilation”). On that definition, ~47% of the raw cohort — 24,056 patients — had at least one confirmed ventilated hour.

Q. Aren’t your exclusions biasing toward sicker, longer-ventilated patients? *[tough]*

A. Fair, and it’s a real limitation we acknowledge. The cohort skews toward sustained, well-documented ventilation courses — that’s the price of needing reliable hourly data to compute the exposure at all; you can’t measure what was never recorded. The exclusions are transparent and pre-specified, so the population we generalize to is clearly defined: adults on a sustained course of invasive ventilation.

Q. How do you know QC didn’t throw away real physiology? *[tough]*

A. The bounds were deliberately wide and physiology-based (e.g., PEEP 0–50, tidal volume 100–2,000 mL, rate ≤ 80 , peak ≤ 100) — outside those are recording artifacts, not human readings. We treated them as missing, and it was tiny: only 1,606 cells across millions. The 22 and 101 patients dropped later were cases where the entire MP column or the analytic variables couldn’t be computed at all — nothing to analyze, not values we disliked.

3) Mechanical power — computation

Q. What units, and what does it represent? *[basic]*

A. Joules per minute — an energy rate: how much mechanical energy the ventilator delivers into the lungs each minute. Higher = more energy into lung tissue per minute.

Q. Which variables go into it? *[basic]*

A. Four routinely charted values: respiratory rate, tidal volume, peak airway pressure, and PEEP — pulled hourly, computed only when all four were present.

Q. Why peak pressure instead of plateau? *[intermediate]*

A. Plateau requires a special inspiratory-hold maneuver that's almost never charted routinely; peak is measured automatically on every breath. The simplified dynamic Gattinoni equation is built around peak precisely so it's computable retrospectively — that trade-off is what makes a study this size possible.

Q. Walk me through the equation. *[intermediate]*

A. $MP = 0.098 \times \text{respiratory rate} \times \text{tidal volume (in liters)} \times [\text{peak pressure} - \frac{1}{2}(\text{peak} - \text{PEEP})]$. We convert tidal volume mL→L first; the bracket estimates the average distending pressure across the breath; 0.098 lands the answer in joules/min.

Q. How do you know the surrogate is trustworthy? *[tough]*

A. We didn't invent it — it's published and validated against the gold-standard pressure-volume reference method by Chiumello and colleagues. We're relying on a peer-validated surrogate, with the accepted trade-off that it lets us compute MP from routine records at scale.

Q. What range/mean did you see — is it sensible? *[tough]*

A. ~0.24 to 109.6 J/min, mean ~13.45 — in the expected ballpark for ventilated ICU patients, with a wide spread reflecting the real mix of lung conditions. We also ran physiological bounds on the inputs first, so charting errors were screened out beforehand.

4) Exposure & time-weighting

Q. Why three exposure metrics? *[basic]*

A. Each captures a different facet and together they stress-test the result: the time-weighted average is the typical intensity, the daily median is a robustness check against spikes, and total cumulative joules is overall workload. If all three agree, we're far more confident it isn't an artifact of how we defined exposure.

Q. Why time-weight the average? *[basic]*

A. Because a plain average would mislead. Settings are charted on change, not at fixed clock times, so each row covers an unequal stretch. A simple mean would count a setting held 1 hour the same as one held 10. We multiply each hour's MP by how long it was actually held, sum, and divide by total time — so it reflects energy actually delivered, not how often the chart was updated.

Q. What does the median add? *[intermediate]*

A. Outlier protection. Coughing or suctioning causes brief MP spikes that can pull an average up; the median reflects the sustained central level. If the finding holds with both, it isn't driven by a few short bursts.

Q. What's immortal-time bias, and how do you avoid it? *[intermediate]*

A. It's where you accidentally credit patients with survival time during a window when they had to be alive for something to be measured. If cumulative power accumulated over the whole stay, longer survivors would rack up more power just by living longer. We avoid it by locking exposure to the first 24 hours and starting the outcome clock at the exact intubation time, not ICU admission — so everyone contributes the same fixed window.

Q. Why only the first 24 hours? *[tough]*

A. It gives a clean, fixed, comparable window for every patient — essential for avoiding immortal-time bias — and it captures the initial ventilation strategy, when many lung-protective decisions are made. We

acknowledge as a limitation that we don't capture prolonged ventilation; that's a question for a future time-varying analysis.

5) Covariates & statistics

Q. Which software did what? *[basic]*

A. SQL Server did the heavy raw-data work — the whole funnel from ~110 million rows down. SPSS built the per-patient variables and ran all statistics. Excel managed the final dataset; Python in Google Colab drew the forest plot. SQL/analysis scripts were drafted with AI assistance, with every output reviewed by us.

Q. What did you adjust for, and why? *[basic]*

A. Age, sex, predicted body weight, an oxygenation ratio, and a pulmonary phenotype from five diagnosis flags (ARDS, pneumonia, COPD, asthma, heart failure) — plus two non-pulmonary severity markers. The goal was to isolate mechanical power's effect, not just "being sicker."

Q. Why vasopressors and CRRT specifically? *[intermediate]*

A. They're objective proxies for severity in other organ systems: vasopressors flag circulatory shock, CRRT (continuous dialysis) flags severe kidney injury. Sicker patients get both more energy and these therapies, so adjusting prevents crediting MP for deaths really driven by shock or kidney failure.

Q. Why not the SOFA score? *[intermediate]*

A. We'd have liked to, but SOFA had too many missing values — using it would have shrunk the sample substantially. So we used well-recorded objective proxies (vasopressors, CRRT) for the same concept. A deliberate trade-off to protect sample size.

Q. How did you build oxygenation / handle missing PBW? *[intermediate]*

A. Preferred $\text{PaO}_2/\text{FiO}_2$ from the worst arterial blood gas in the first 24h; where no gas existed, we substituted the validated $\text{SpO}_2/\text{FiO}_2$ so we didn't lose patients. Missing PBW (a small number) was imputed with the sex-specific median.

Q. Why logistic regression, and three parallel models? *[tough]*

A. The outcome is binary (alive/dead at 28 days), so logistic regression fits and gives covariate-adjusted odds ratios. The three MP summaries are collinear — you can't put them in one model — so we ran three parallel models with identical covariates to confirm the finding holds however we summarize exposure.

Q. What does the interaction term test? *[tough]*

A. Subgroup models alone give the effect within each stratum. An interaction term formally tests whether the effect actually differs between strata (e.g., under-65 vs 65+) rather than by chance — the specific numbers belong to the results section.

6) Limitations / skeptical questions

Q. Single Boston hospital — does it apply elsewhere? *[basic]*

A. Fair, and we're upfront. Case mix, staffing, and ventilation practices may differ. But the physics of mechanical power — energy per minute into the lung — isn't hospital-specific, so the signal is plausibly transportable even if exact numbers aren't. External, ideally multi-center validation is the honest next step.

Q. Observational — association, not causation? *[basic]*

A. Correct, and we say exactly that. We show an association; we can't prove causation from this design. We did what observational studies can — multivariable adjustment, pre-specified subgroups, fixed exposure window — but the word is association. Causation needs a trial randomizing a power-limiting strategy.

Q. Aren't you just measuring severity, not harm from MP? *[intermediate]*

A. That's confounding by indication, the central threat, and we tackled it directly — adjusting for age, sex, oxygenation, vasopressors (shock), and dialysis (kidney injury). But adjustment only handles measured severity; residual confounding is a real limitation. So our claim is an independent association after adjustment, not a clean separation.

Q. Doesn't peak pressure overestimate energy with high airway resistance? *[intermediate]*

A. You've found the key trade-off. Peak includes a resistive component, so in high-resistance patients it can overestimate energy reaching the lung versus a plateau-based calculation. We accept it because peak is routinely recorded and needs no inspiratory-hold maneuver — the only thing that makes MP computable retrospectively across millions of hours. It's not perfect, but the dynamic surrogate is validated against the gold-standard reference method.

Q. ICD codes are imprecise — how much could that distort things? *[intermediate]*

A. Agreed — they're billing codes, not clinical adjudication, so some phenotypes are mislabeled. But here they're covariates we adjust for, not the main exposure, so misclassification mainly weakens our lung-phenotype adjustment rather than directly biasing the power estimate. A prospective study with confirmed phenotypes would tighten this.

Q. You left out sedation, spontaneous effort, and SOFA — doesn't that undermine adjustment? *[tough]*

A. Real gaps, and we won't pretend otherwise. We lacked reliable data on sedation depth and spontaneous breathing effort — both affect transmitted energy — so they sit in our unmeasured-confounding bucket, and SOFA was dropped for missingness. The honest bottom line: residual confounding remains, which is exactly why we frame this as an association and call for prospective validation.

4. Accuracy Fixes Baked In

- “run by MIT” → corrected so MIT is credited with building the database, not running the hospital.
- “freely available” → “freely accessible to credentialed researchers” — accurate to the access model, and it pre-empts an examiner challenge.